Numbers 0-20

QUICK REFERENCE | Suggested grades K-1 | Math

Say it. Find it. Count one object for each number.

Number names

0 zero	1 one	2 two
3 three	4 four	5 five
6 six	7 seven	8 eight
9 nine	10 ten	11 eleven
12 twelve	13 thirteen	14 fourteen
15 fifteen	16 sixteen	17 seventeen
18 eighteen	19 nineteen	20 twenty

Count in order

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

11, 12, 13, 14, 15, 16, 17, 18, 19, 20

One more / one less

One more than 8 is 9. One less than 8 is 7.

Compare two groups

More means a larger amount. Fewer means a smaller amount.

The same number means equal. Match objects one to one to compare.

OPP

Place value

QUICK REFERENCE | Suggested grades 1-4 | Math

A digit gets its value from its place in a number.

Read 4,582

Thousands	Hundreds	Tens	Ones
4	5	8	2
4,000	500	80	2

Three ways to write it

Standard: 4,582

Expanded: 4,000 + 500 + 80 + 2

Word form: four thousand five hundred eighty-two

Trade without changing the value

10 ones = 1 ten. 10 tens = 1 hundred.

10 hundreds = 1 thousand.

Compare whole numbers

Start at the left. Compare the first place with different digits.

4,582 > 4,528 because 8 tens is more than 2 tens.

Round a whole number

Find the place. Look one place to its right.

0-4: keep the digit. 5-9: add 1 to the digit.

Change all digits to the right to zero. Example: 4,582 to the nearest hundred is 4,600.

OPP

Addition & subtraction

QUICK REFERENCE | Suggested grades K-2 | Math

Addition joins amounts. Subtraction takes away or finds the difference.

Count on to add

Start with 7. Add 3: say 8, 9, 10. So 7 + 3 = 10.

Make a ten

8 + 5 = 8 + 2 + 3 = 10 + 3 = 13.

Use a double

6 + 6 = 12, so 6 + 7 is one more: 13.

Think addition to subtract

For 13 - 8, ask: 8 plus what makes 13?

8 + 5 = 13, so 13 - 8 = 5.

A fact family uses the same three numbers

Addition	Subtraction
4 + 6 = 10	10 - 4 = 6
6 + 4 = 10	10 - 6 = 4

Check your answer

Use objects, draw dots, or use the opposite operation.

For 9 - 4 = 5, check that 5 + 4 = 9.

OPP

Multiplication chart 0-12

QUICK REFERENCE | Suggested grades 2-6 | Math

Find one factor on the left and the other across the top. Meet at the product.

x	0	1	2	3	4	5	6	7	8	9	10	11	12
0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	10	11	12
2	0	2	4	6	8	10	12	14	16	18	20	22	24
3	0	3	6	9	12	15	18	21	24	27	30	33	36
4	0	4	8	12	16	20	24	28	32	36	40	44	48
5	0	5	10	15	20	25	30	35	40	45	50	55	60
6	0	6	12	18	24	30	36	42	48	54	60	66	72
7	0	7	14	21	28	35	42	49	56	63	70	77	84
8	0	8	16	24	32	40	48	56	64	72	80	88	96
9	0	9	18	27	36	45	54	63	72	81	90	99	108
10	0	10	20	30	40	50	60	70	80	90	100	110	120
11	0	11	22	33	44	55	66	77	88	99	110	121	132
12	0	12	24	36	48	60	72	84	96	108	120	132	144

OPP

Ways to remember

Any number x 0 = 0. Any number x 1 = itself.

3 x 4 = 4 x 3. Turn an array to see the same total.

6 x 7 = (5 x 7) + (1 x 7) = 35 + 7 = 42.

Connect multiplication and division

4 x 6 = 24, so 24 divided by 6 = 4 and 24 divided by 4 = 6.

Division reminders

QUICK REFERENCE | Suggested grades 3-6 | Math

Division makes equal groups or finds the size of each group.

Think of a related multiplication fact

42 divided by 6 = ? Think: 6 x ? = 42. Answer: 7.

Divide 156 by 3, one place at a time

1. 1 hundred cannot make 3 whole hundreds in each group. Use 15 tens.

2. 15 tens divided by 3 = 5 tens. Write 5 in the tens place.

3. Bring down the 6 ones. 6 divided by 3 = 2 ones.

4. Answer: 52. Check: 52 x 3 = 156.

A remainder is an amount left over

17 divided by 5 = 3 remainder 2, because 5 x 3 + 2 = 17.

The remainder must be smaller than the divisor.

What does the problem ask?

17 people, 5 seats per car: need 4 cars, because everyone needs a seat.

17 stickers shared by 5 students: 3 each, with 2 left over.

Remember

You cannot divide by zero. 0 divided by a nonzero number is 0.

OPP

Fractions, decimals & percents

QUICK REFERENCE | Suggested grades 4-6 | Math

These are different ways to name the same part of a whole.

Common equivalents

Fraction	Decimal	Percent
1/10	0.1	10%
1/8	0.125	12.5%
1/5	0.2	20%
1/4	0.25	25%
1/3	0.333...	33 1/3%
1/2	0.5	50%
2/3	0.666...	66 2/3%
3/4	0.75	75%
1 whole	1.0	100%

OPP

Fraction to decimal

Divide the numerator by the denominator. 3/4 = 3 divided by 4 = 0.75.

Decimal to percent

Multiply by 100. 0.75 x 100 = 75, so 0.75 = 75%.

Percent to decimal

Divide by 100. 25% = 25/100 = 0.25.

Watch the repeating decimals

The dots in 0.333... mean the 3 repeats forever. 0.33 is only an approximation of 1/3.

Working with fractions

QUICK REFERENCE | Suggested grades 3-6 | Math

Use equal-sized parts. The denominator tells the size of the parts.

Name the parts

In 3/4, the numerator 3 counts the parts.

The denominator 4 means the whole is split into 4 equal parts.

Make an equivalent fraction

Multiply or divide the top and bottom by the same nonzero number.

1/2 = 2/4 = 3/6. Divide 6/8 by 2 on top and bottom to get 3/4.

Add or subtract: use a common denominator

1/2 + 1/3 = 3/6 + 2/6 = 5/6.

3/4 - 1/2 = 3/4 - 2/4 = 1/4.

Add or subtract the numerators. Keep the common denominator.

Multiply across

2/3 x 3/4 = (2 x 3)/(3 x 4) = 6/12 = 1/2.

Divide: multiply by the reciprocal

3/4 divided by 1/2 = 3/4 x 2/1 = 6/4 = 1 1/2.

Do not divide by zero.

Check the whole

Compare fractions of the same-sized whole. Draw fraction bars if you need a picture.

OPP

Measurement conversions

QUICK REFERENCE | Suggested grades 3-6 | Math

Use matching units. These customary measures use U.S. units.

Length

U.S. customary	Metric
12 inches = 1 foot	10 millimeters = 1 centimeter
3 feet = 1 yard	100 centimeters = 1 meter
5,280 feet = 1 mile	1,000 meters = 1 kilometer

Capacity and mass / weight

U.S. liquid capacity	Metric capacity
8 fluid ounces = 1 cup	1,000 milliliters = 1 liter
2 cups = 1 pint	Metric mass
2 pints = 1 quart	1,000 milligrams = 1 gram
4 quarts = 1 gallon	1,000 grams = 1 kilogram

U.S. customary weight

16 ounces = 1 pound. 2,000 pounds = 1 short ton.

Choose multiply or divide

Larger unit to smaller unit: multiply. 3 feet x 12 = 36 inches.

Smaller unit to larger unit: divide. 200 centimeters divided by 100 = 2 meters.

Check the label

A fluid ounce measures volume. An ounce of weight measures weight. They are not the same kind of measurement.

Shapes, area & perimeter

QUICK REFERENCE | Suggested grades 2-6 | Math

Perimeter is distance around. Area measures a surface.

Shape facts

Shape	Remember
Triangle	3 sides; angles total 180 degrees
Quadrilateral	4 sides; angles total 360 degrees
Rectangle	4 right angles; opposite sides equal
Square	4 equal sides and 4 right angles
Circle	Radius: center to edge. Diameter = 2 x radius.

Rectangle

Perimeter = 2 x (length + width). Area = length x width.

Example: length 5 cm, width 3 cm. Perimeter = 16 cm. Area = 15 square cm.

Triangle

Area = (base x perpendicular height) divided by 2.

Example: base 8 cm, height 3 cm. Area = 12 square cm.

Rectangular prism

Volume = length x width x height. Use cubic units.

Example: 4 cm x 3 cm x 2 cm = 24 cubic cm.

Label your answer

Length: cm. Area: square cm. Volume: cubic cm.

OPP

Decimals & operation order

QUICK REFERENCE | Suggested grades 4-6 | Math

Read places carefully and estimate before calculating.

Read 3.456

Ones	Tenths	Hundredths	Thousandths
3	4	5	6

Add and subtract

Line up decimal points. Add zeros to show equal places.

2.40 + 0.35 = 2.75. 5.00 - 1.28 = 3.72.

Multiply decimals

Multiply as whole numbers. Count decimal places in BOTH factors.

1.2 x 0.3: 12 x 3 = 36. Two decimal places total gives 0.36.

Divide by a decimal

Multiply BOTH numbers by the same power of 10 to make the divisor whole.

4.8 divided by 0.6 = 48 divided by 6 = 8.

Order of operations

1. Grouping symbols, such as parentheses. 2. Exponents.

3. Multiply and divide from left to right.

4. Add and subtract from left to right.

Example: 18 - 6 divided by 3 x 2 = 18 - 2 x 2 = 18 - 4 = 14.

OPP

Time & money

QUICK REFERENCE | Suggested grades 1-3 | Math

Take one step at a time. Label minutes, hours, cents, or dollars.

U.S. coins

Penny	Nickel	Dime	Quarter
1 cent	5 cents	10 cents	25 cents

Count money

Start with the largest coin value and count on.

1 quarter + 2 dimes + 1 nickel = 25 + 10 + 10 + 5 = 50 cents.

100 cents = $1.00. $2.35 means 2 dollars and 35 cents.

Read an analog clock

The short hand tells the hour. The long hand tells minutes.

Each number marks 5 minutes: 1 = 5, 2 = 10, 3 = 15 ... 12 = 0.

At 3:30, the minute hand points to 6; the hour hand is halfway from 3 to 4.

Time facts

60 seconds = 1 minute. 60 minutes = 1 hour. 24 hours = 1 day.

A quarter hour is 15 minutes. A half hour is 30 minutes.

Find elapsed time

From 2:45 to 3:20: 15 minutes to 3:00, then 20 minutes more.

15 + 20 = 35 minutes. Use a timeline to track the jumps.

OPP

Word problems: a calm plan

QUICK REFERENCE | Suggested grades 1-6 | Math

You do not have to solve it all at once. Start with what you know.

1. Read and tell the story

Read once. Say what is happening in your own words.

Circle the question. What are you trying to find?

2. Find the useful information

Label each amount. Cross out information you do not need.

Draw objects, a bar model, a table, or a number line.

3. Choose a plan

Are amounts joining, separating, being compared, or forming equal groups?

Write an equation. Do not choose an operation from one keyword alone.

4. Solve one step at a time

Show your work. Use a chart or a familiar fact if you need one.

5. Check and label

Does your answer fit the story? Check with a drawing or another method.

Write a sentence with the correct unit.

Example

Mia has 3 bags with 4 shells in each. She gives away 2 shells.

3 x 4 = 12 shells. 12 - 2 = 10 shells. Mia has 10 shells left.

OPP

Nouns: name it

QUICK REFERENCE | Suggested grades K-3 | English

A noun names a person, place, thing, or idea.

Find the noun

Person	Place	Thing	Idea
teacher	school	pencil	kindness
friend	park	dog	courage

Common or proper?

A common noun names a general person, place, thing, or idea: city, student.

A proper noun names a specific one: Chicago, Maya. Use a capital letter.

One or more than one

Singular: one cat. Plural: two cats.

Usually add -s: book/books. Often add -es: box/boxes.

Consonant + y: baby/babies. Some change: child/children, mouse/mice.

Show ownership

One dog: the dog's bowl. More than one dog: the dogs' bowls.

Irregular plural: the children's books.

Try it

The teacher carries a book to school.

Nouns: teacher (person), book (thing), school (place).

OPP

Verbs: action & being

QUICK REFERENCE | Suggested grades 1-4 | English

A verb tells what happens or connects the subject to information about it.

Action verbs

A penguin waddles. The class reads. We built a tower.

Waddles, reads, and built tell what someone or something does.

Linking verbs

The soup is warm. The flowers smell sweet.

Is and smell connect the subject to a description.

Helping verbs

She is reading. They have finished. We will go.

Is, have, and will help the main verb.

Past, present, future

Past	Present	Future
Yesterday I walked.	Today I walk.	Tomorrow I will walk.
Yesterday I went.	Today I go.	Tomorrow I will go.

Match subject and verb

One bird sings. Two birds sing.

I am. You are. He/she/it is. We are. They are.

Try it

Lena has packed her bag. Helping verb: has. Main verb: packed.

OPP

Adjectives & adverbs

QUICK REFERENCE | Suggested grades 2-5 | English

Use describing words to make a sentence more precise.

Adjectives describe nouns or pronouns

Ask: What kind? Which one? How many?

The small penguin found three smooth stones.

Small describes penguin. Three and smooth describe stones.

Adverbs describe verbs, adjectives, or other adverbs

Ask: How? When? Where? To what extent?

The penguin waddles slowly. Slowly tells how it waddles.

The water is very cold. Very describes the adjective cold.

Compare carefully

One	Compare two	Compare three or more
small	smaller	smallest
careful	more careful	most careful
good	better	best

Do not rely only on -ly

Quickly is an adverb, but friendly is usually an adjective.

Fast can be either: a fast runner (adjective); runs fast (adverb).

Quick check

Ask which word is being described. Then choose the word that fits.

Pronouns & clear sentences

QUICK REFERENCE | Suggested grades 1-4 | English

A pronoun can take the place of a noun. Make it clear who or what it means.

Choose a pronoun

Subject: does the action	Object: receives the action
I, you, he, she, it, we, they	me, you, him, her, it, us, them

Subject or object?

Maya and I read. I is part of the subject.

The teacher helped Maya and me. Me is part of the object.

Check by removing the other name: I read. The teacher helped me.

Possessive words show belonging

My/your/his/her/its/our/their come before nouns: our books.

Mine/yours/his/hers/ours/theirs can stand alone: The books are ours.

Its or it's?

Its shows belonging: The bird spread its wings.

It's means it is or it has: It's raining.

Make the meaning clear

Unclear: Sam told Lee that he won. Which person won?

Clear: Sam told Lee, "You won." Or: Sam told Lee, "I won."

OPP

Sentences & punctuation

QUICK REFERENCE | Suggested grades K-3 | English

A complete sentence shares a complete thought.

Build a sentence

Tell who or what (subject). Tell what happens or what is true (predicate).

The little dog / sleeps on the rug.

Begin with a capital letter. End with a punctuation mark.

Choose an ending

Type	Example
Statement: tells something	The dog is asleep.
Question: asks something	Where is the dog?
Exclamation: strong feeling	What a surprise!
Command: tells someone to act	Please close the door.

Commas can separate a list

I packed a book, a snack, and a pencil.

Apostrophes have jobs

Contraction: do not becomes don't.

Ownership: the cat's toy. Do not add an apostrophe just to make a plural.

Read it aloud

Does it make sense? Is the thought finished? Check the first and last marks.

OPP

My writing check

QUICK REFERENCE | Suggested grades 2-6 | English

Read your writing one small part at a time.

1. Meaning

What is my main idea? Did I answer the question?

Add a detail or example to help the reader understand.

2. Order

Does my beginning introduce the idea?

Do my details follow an order? Does my ending finish the thought?

3. Sentences

Read each sentence aloud or ask someone to read it to you.

Does each sentence make sense? Join or separate thoughts where needed.

4. Capitals and punctuation

Check the first letter, names, and the word I.

Check sentence endings, commas, and apostrophes.

5. Spelling and word choice

Mark words you are unsure about. Use a word bank or dictionary.

Check words that sound alike: their/there/they're and to/too/two.

6. One final look

Check spaces, readable handwriting or text, and your name if needed.

Choose one improvement. A small change can make your message clearer.

OPP

Connecting ideas

QUICK REFERENCE | Suggested grades 4-6 | English

Use connecting words to show how ideas belong together.

Coordinating conjunctions

For, and, nor, but, or, yet, so can join words, phrases, or clauses.

I wanted to play, but it was raining.

Use a comma before the conjunction when it joins two complete sentences.

Subordinating conjunctions

Because, although, if, when, and while begin dependent clauses.

Because it was raining, we played inside.

We played inside because it was raining.

A dependent clause does not usually stand alone as a complete sentence.

Prepositions show relationships

In, on, under, beside, before, after, and with can begin prepositional phrases.

The book is under the desk. We walked after lunch.

Transitions guide the reader

Order: first, next, finally. Example: for instance.

Contrast: however. Cause/result: therefore. Add information: also.

Check the connection

Choose a connector that matches your meaning. Read the sentence to check that the ideas still fit.

OPP